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Ans: to the Question no: 1

(a-i) Image processing:- Image processing is the technique of performing operations on an image to enhance it, extract useful information, or prepare it for analysis, such as filtering, compression, segmentation, or recognition.

(a-ii) Noise:- Noise is any unwanted or random variation in pixel intensity values that degrades the quality of an image.

(a-iii) Zero Crossing:- A zero crossing occurs when the sign of a function changes from positive to negative or vice-versa. In image processing, it is used to detect edges.

(2)

(Q-iv) Wavelet:- A wavelet is mathematical function used for multi-resolution analysis, representing data at different frequency components with spatial localization. It is widely used in compression, denoising etc

b) Illumination Vs Reflection

Illumination ($I(x,y)$)

- The amount of light falling on the scene from an external source.
- It depends on lighting condition
- Varies slowly across the image

Reflectance ($R(x,y)$)

- The fraction of incident light reflected by the object's surface.

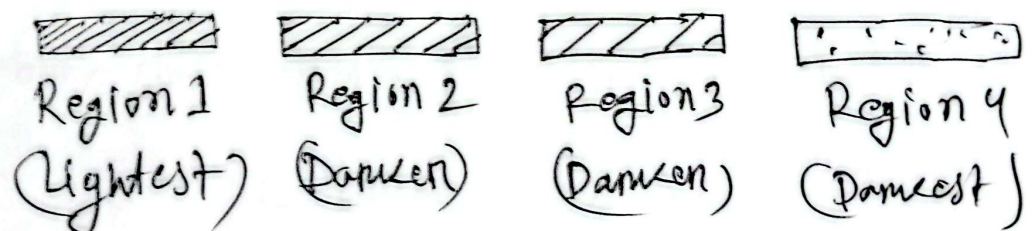
- It is property of the material
- varies abruptly at object boundaries

c) Match Band Effect

It is an optical illusion where human eye perceives exaggerated contrast at edges of slightly different shades of gray.

Sketch:

Suppose we have 4 regions of gray, each slightly darker than the previous.



Although the actual intensity changes smoothly, our brain enhances the edge, creating bright and dark bands that don't really exist.

d) Application of Digital Image processing :-

1) Medical imaging → MRI, CT scan, X-rays,

2) Remote Sensing → Satellite image analysis,
Weather monitoring

3) Industrial inspection → Detecting defects
in products

4) Security → Face recognition, biometric auth.

5) Robotics & AI → Object detection, enhancement
for evidence

6) Forensics → Image restoration

e) i) Color map :- A color map is a lookup
table that maps pixel intensity values
to specific colors for visualization
and analysis -

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ii) FFT (Fast Fourier Transform):- FFT is an efficient algorithm to compute the DFT, which converts an image from the spatial domain to the frequency domain.

iii) Bit-plan:- A bit plan represents an image using the binary values of pixels. Each pixel's intensity is split into bits where higher bitplans capture major structures and lower ones capture fine detailed or noise.

iv) Discrete Laplacian:- The discrete Laplacian is a second-order derivative operator used in image processing to highlight regions of rapid intensity change.

Example mask:-

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

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f) What is image histogram?

An image histogram is a graphical representation of the distribution of pixel intensity values in an image

- On the x-axis $\rightarrow$ intensity values
- On the y-axis $\rightarrow$ number of pixels

Types

1. Gray Scale Image Histogram $\rightarrow$ shows how many pixel exist at each gray level.
2. Color Image Histogram :- Separate histograms for R, G, B channel-

Usefulness :-

- 1) Describe Overall brightness and contrast
- 2) Helps in image enhancement
- 3) Useful for segmentation, thresholding and image analysis.

Ans! to the question number 2

a) SIFT (Scale-Invariant Feature Transform)

Purpose:- Detects and describes local features in images that are invariant to scale, rotation, and illumination changes

Steps:

- 1) Scale-space extrema detection → Finds keypoints using difference of Gaussian at multiple scales
- 2) Keypoint Localization → Filters out weak keypoints
- 3) Orientation assignment → Assignment a dominant orientation to each keypoint
- 4) Keypoint descriptor:- Around each keypoint a 128-dimensional vector is formed using gradient orientations -

SURF (Speeded-UP Robust Features)

⊗ purpose: Similar to SIFT, but designed to be faster

⊗ Steps:-

- 1) Uses Hessian matrix (based on second-order derivative) to detect keypoints
- 2) Approximate Gaussian derivative using box filters + integral images
- 3) Keypoint description are created using Haar wavelet response around the neighbourhood.

b) Different Energy Sources Used in image creation:-

Images in digital image processing are typically formed by capturing energy reflect or emitted from objects. The most common energy source include!

1. Electromagnetic Energy (light) :-

- ⊗ Visibility light is the most common source for everyday image
- ⊗ Works by capturing reflected or transmitted light from objects
- ⊗ Example: Mobile phone camera, optical microscope

2. X-Rays :-

- ⊗ X-Ray pass through soft materials but are absorbed by dense structures
- ⊗ Very useful in meaning medical imaging and security scan.
- ⊗ Example: X-ray radiography

3. Gamma Rays :-

- ⊗ Even higher energy than X-rays

C) Importance of Gaussian filters

1. Noise Reduction

- ⊗ Gaussian filter is widely used to remove noise from image
- ⊗ It smoothes the images by averaging pixel values with a weighted G.D.:-

2. Edge preservation:

- ⊗ Unlike uniform filters, Gaussian filters reduce high-frequency noise but preserve edge better
- ⊗ This is important in application like edge detection where noise can cause false edges-

3. Mathematical properties

A 2D Gaussian filter can be applied

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- ⊗ Used in nuclear medicine imaging
 - ⊗ Helps visualize metabolic process inside the body

4. Infrared Radiation :-

- ⊗ Capture thermal emission from objects
- ⊗ Useful for night vision, remote sensing, and heat detection
- ⊗ Example: Thermal Camera

5. Ultrasound

- ⊗ Use high frequency sound waves
- ⊗ Reflections of sound are captured and converted into images
- ⊗ Example : Ultrasound imaging in medical diagnosis.

6. Radio Waves / Microwaves

⊗ Used in Radar and MRI

⊗ MRI uses Strong magnetic fields and radio waves to form detailed images of soft tissues.

⊗ Radar imaging helps in weather predicts and remote sensing

7. Electron Beams

⊗ Instead of light, electron are used to create very high-resolution images

⊗ Example:

1) TEM

2) SEM

as two 1D filters, reduce computational cost -

⊗ It is also rotationally symmetric, meaning smoothing is applied equally in all direction

4. Frequency Domain Advantages

⊗ Gaussian filtering corresponding to a low-pass filter in frequency domain

⊗ This help in removing high-frequency components while keeping low-frequency

5.) preprocessing for computer vision:

Gaussian filters are often used as a pre-processing step before

⊗ Edge detection

⊗ Image pyramids

⊗ Segmentation

⊗ Feature extraction

Ans: to the Question number 3

(a-i) import cv2

from matplotlib import pyplot as plt

img = cv2.imread("img-path")

channels = ('b', 'g', 'r')

plt.figure(figsize=(10, 5))

plt.title("Histogram of RGB image")

plt.xlabel("pixel intensity")

plt.ylabel("frequency")

for i, col in enumerate(channels):

hist = cv2.calcHist([img], [i], None, [256],
[0, 256])

plt.plot(hist, color=col)

plt.xlim([0, 256])

plt.show()

a-ii) import cv2

from matplotlib import pyplot as plt

img = cv2.imread("img-path", cv2.IMREAD_GRAYSCALE)

avg-filtered = cv2.blur(img, (5,5))

cv2.imshow("Original Gray Image", img)

cv2.imshow("Average filtered image", avg-filtered)

cv2.waitKey(0)

cv2.destroyAllWindows()

a-iii) import cv2

import numpy as np

img = cv2.imread("img-path", cv2.IMREAD_GRAYSCALE)

def saltAndpepper-noise(img, prob=0.02):

noisy = np.copy(img)

numSalt = np.ceil(prob * img.size * 0.5)

coords = [np.random.randint(0, i-1, int(

numSalt)) for i in noisy[coords[0]]

coords[1]] = 0

return noisy

```

noisy_img = add_salt_and_pepper_noise(img)
median_filtered = cv2.medianBlur(noisy_img, 3)
cv2.imshow("Original Gray Image", img)
cv2.imshow("Noisy salt-and-pepper-img", noisy_img)
cv2.imshow("Median Filtered img", median_filtered)
cv2.waitKey(0)
cv2.destroyAllWindows()

```

b-i) Difference between brightness and contrast

Aspect	Brightness	Contrast
Definition	Refers to overall lightness or darkness of an image	Refers to the difference in intensity between the light and dark areas
Effects	Shift all pixels values up or down equally, making the image lighter or darker	Stretches or compresses the range of pixel values
Analogy	Like turning up/down a light dimmer switch in a room	Like adjusting the sharpness of shadows between bright and dark objects
Pixel Behavior	Adds or subtracts a constant value to all pixel intensity	Expands or shrinks the spread between minimum and max values
Use case	Used when an image is overall too dark or too bright	Used to highlight details and edges, making features stand out.

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b-i) Convolution property: States that in the frequency domain, convolution in the spatial domain becomes multiplication, and multiplication in the spatial domain become convolution.

Mathematically

⊗ If two 2D signals/images $f(x,y)$ and $h(x,y)$ are convolved:

$$g(x,y) = f(x,y) \otimes h(x,y)$$

Where $\otimes$ denotes convolution

Then their 2D Fourier Transform satisfies

$$G(u,v) = F(u,v) \cdot H(u,v)$$

That's mean;

Convolution in spatial domain $\leftrightarrow$ multiplication in frequency domain.

What it means in image processing:

Filtering: Many image operations are convolutions instead of convolving in the spatial domain, we can multiply in the frequency domain after applying the 2D Fourier Transform.

Example in image processing

- 1) Suppose $f(x,y)$ = image
- 2) $h(x,y)$ = blur kernel
- 3) Convolution in spatial domain $\rightarrow$ blur image

$$g(x,y) = f(x,y) * h(x,y)$$
- 4) In frequency domain:
 - i) Take Fourier transform: $F(u,v), H(u,v)$
 - ii) Multiply: $G(u,v) = F(u,v) * H(u,v)$
 - iii) Take inverse Fourier transform $\rightarrow g(x,y)$ (blurred)

C-i) Difference between Linear and non linear filter

Linear filter:

Definition: A filter is linear if the output is a linear combination of the input pixels.

- ⊗ They satisfy superposition principle
- ⊗ Usually implemented with convolution.

Example: 1) Mean filter

2) Gaussian

3) Laplacian

4) Sobel, Prewitt

Properties: 1) Simple to implement mathematically
2) Good for smoothing and edge detection
3) But sensitive to noise, especially impulse noise.

Non-linear Filters:

Definition: A filter is non-linear if the output pixel is not linear combination of input pixel

- ⊗ They don't satisfy superposition principle
- ⊗ often involve ranking or non-linear function on neighbourhood pixels,

Example: 1) median filter

- 2) Max/min filter
- 3) Bilateral filter
- 4) Morphological filter

- properties:
- 1) Better at removing non-Gaussian noise
 - 2) Can preserve edge and details while denoising
 - 3) More computationally complex compared to linear filter-

C-ii) Importance of Hue and Saturation in Color image formation:-

A Color image in models like HSI/HSV is represented by Hue (H), Saturation (S) and Intensity/Values (I/V)

Hue: Represents types of color (red, green, blue)

⊗ It corresponds to the wavelength of light

Example:

Hue $\approx 0 \rightarrow$ Red
 Hue $\approx 120 \rightarrow$ Green
 Hue $\approx 240 \rightarrow$ Blue

Importance :

- i) Distinguishes different pure colors
- ii) Determine the preceived color in the image
- iii) Used in segmentation, object recognition and tracking

Saturation (S) : Represents the purity or vividness of the color

- ⊗ High saturation = pure, vibrant color
- ⊗ Low saturation = dull, washed-out, grayish color

Importance :

- 1) Controls color richness and strength
- 2) Helps in enhancing or suppressing colors in image editing
- 3) Useful in image enhancement, medical imaging, remote sensing.

d) General Model of image degradation

Image degradation refers to the distortion of an ideal image due to blurring and noise during acquisition, transmission or storage. A mathematical model helps to understand and reverse this degradation

Mathematical Representation

Spatial domain model:

$$g(x,y) = f(x,y) * h(x,y) + n(x,y)$$

Where:

$f(x,y)$ = original image

$h(x,y)$ = Degradation filter

$*$ = Convolution operation

$n(x,y)$ = Additive noise

$g(x,y)$ = Degraded image

Frequency Domain model:

To recover the original image:

$$F(u,v) = (G(u,v) - N(u,v)) / H(u,v)$$

issue:

- i) Exact noise $N(u,v)$ is unknown
- ii) If $H(u,v)$ is close to zero, division cause instability.

Step-by-step interpretation

- 1) Original scene: → The real world objects produces the ideal image ($f(x,y)$)
- 2) Degradation function ($h(x,y)$):
 - 1) Motion blur
 - 2) Atmospheric blur
 - 3) Lens imperfection
- 3) Convolution process: → The blur is smeared over pixel neighborhood
- 4) Noise addition ($n(x,y)$):
 - 1) Sensor imperfection
 - 2) Transmission error
 - 3) Quantization

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Ans! to the Question number 4

a) Image compression reduce the size of image files by removing redundant or unnecessary data while keeping acceptable image quality?
It is broadly classified into:

1) Lossless Compression

- i) No information is lost $\rightarrow$ original image can be reconstructed exactly
- ii) Used where accuracy is important

2) Lossy Compression:

- i) Some information is lost, but file size is greatly reduced
- ii) Focuses on human visual perception

3) Transform Coding:

- i) Image is transformed into another domain

4) predictive Coding:

i) uses correlation between neighbouring pixels to predict values and encode the difference

Huffman Coding: Huffman coding is a lossless compression method. It assigns shorter codes to more frequently symbols and longer code to less frequent symbols.

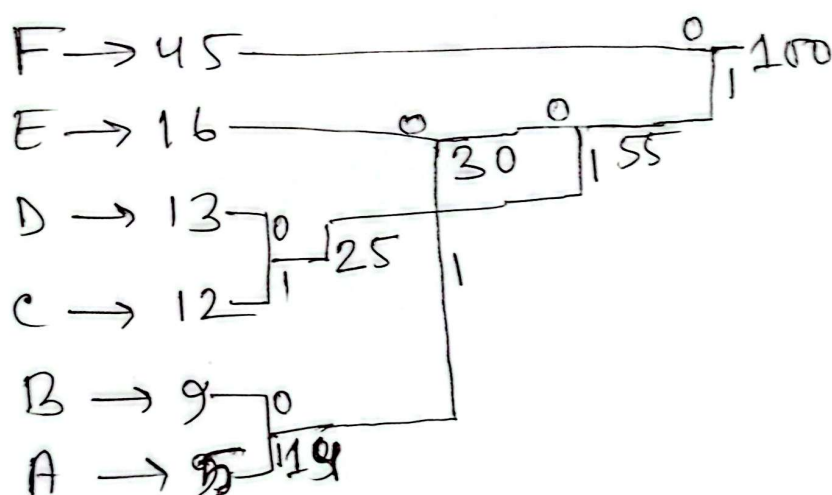
Step of Huffman Coding:

- 1) Count the frequency of each symbol
- 2) Build a binary tree, starting with least frequent symbols.
- 3) Assign 0 and 1 along tree branches
- 4) Generate prefix code
- 5) Replace each symbols with its code-

Example:

Suppose we have symbols and their frequencies

Symbol	Frequency
A	5
B	9
C	12
D	13
E	16
F	45



Final Code Table

Symbol	code
A	1111
B	.1110
C	111
D	110
E	100
F	0

b) ROI (Region of interest filtering)

What is ROI? ROI refers to a specific portion of an image that we want to analyze enhance, or process.

⊗ Instead of applying an operation to the whole image, we restrict it to the selected region only.

Why ROI filtering:-

- 1) Focus on important areas
- 2) Ignore irrelevant regions
- 3) Efficient processing
- 4) Better accuracy

Steps in ROI filtering

1. select/define ROI:

- i) Manually
- ii) Automatically

2. Apply filter only inside ROI
 - i) common filters: Average, Gaussian, median, Sharpening
3. Replace filtered ROI back into original image
 - i) the rest of the image remain unchanged

Example: ROI filter processing

Suppose we want to blur only the face region in an image

- 1) Load image
- 2) Define ROI
- 3) Apply Gaussian blur
- 4) put blurred ROI back into original image

Application of ROI Filtering

- i) Medical imaging
- ii) Surveillance
- iii) Document processing
- iv) Face processing
- v) Object detection

c) Two-Dimensional Moment M_{pq}

Definition: Measures the raw spatial distribution of pixel intensity in an image, weighted by the $x^p y^q$ term.

$$M_{pq} = \sum_x \sum_y x^p y^q f(x, y)$$

Where:

$f(x, y)$ = intensity at pixel (x, y)

p, q = non-negative integers

Key point: i) Dependent on image position
ii) Shifts in the image will change M_{pq}

Central Moment μ_{pq} :

Definition: Like a 2D moment, but computed relative to the image's centroid $(\bar{x}, \bar{y})$ making it translation invariant.

$$\bar{x} = \frac{M_{10}}{M_{00}}, \quad \bar{y} = \frac{M_{01}}{M_{00}}$$

$$\mu_{pq} = \sum_x \sum_y (x - \bar{x})^p (y - \bar{y})^q f(x, y)$$

Key point:

- i) Removing the effect of location
- ii) If you have the object in the image μ_{pq} stays same.

Normalized Central Moment: η_{pq}

Definition: A central moment scaled to be independent of both translation and scale

$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^{\gamma}}, \quad \gamma = \frac{p+q}{2} + 1$$

Key point:

- i) Useful for shape comparison,
- ii) Since it won't change if you move or resize the object.

Ans. to the Question number 5

a) single threshold VS Double threshold

Aspect	Single Threshold	Double Threshold
Number of Threshold	Use one threshold value (T)	Use two threshold value (T_1) and T_2
Condition for Pixel classification	Pixel becomes white if its gray level $> T$ Otherwise black	Pixel becomes white if its gray level lies between (T_1) and T_2 Otherwise black
Purpose	Separates objects from background based on a single cut-off	Isolate objects within a specific range of intensities
Output	Binary Image with broad segmentation	Binary Image focusing on feature in a narrower intensity band
Use case	Simple object detection when foreground and background are easily separable	Edge / boundary detection, highlight structures with specific intensity range -

Application of thresholding

1) Image Segmentation:

i) Thresholding is widely used to separate objects from the background

2) Document Binarization

Thresholding is essential in converting gray scale scanned documents into binary image for OCR

3) Medical Imaging

Thresholding helps in identifying region of interest in medical scans like MRI

4) Background Subtraction in video Surveillance

5) Chemical and biological image Analysis

```

c) import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('img-path')

img-rgb = cv2.cvtColor(img, cv2.COLOR_BGR
                    2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2
                    GRAY)

edge-canny = cv2.canny(gray, 100, 200)

Sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize
                    =3)

Sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

Sobel-Combined = cv2.magnitude(Sobelx, Sobely)

Laplacian = cv2.Laplacian(gray, cv2.CV_64F)

fig, axes = plt.subplots(2, 3, figsize=(14, 8))

axes[0, 0].imshow(img-rgb)
axes[0, 1].imshow(gray, cmap='gray')
axes[0, 2].imshow(edge-canny, cmap='gray')
axes[1, 0].imshow(Sobel-Combined, cmap='gray')
axes[1, 1].imshow(Laplacian, cmap='gray')

```